

Engineering Curves: Construction of Hypocycloid

Rolling Circle Method with Tangent & Normal

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Problem Statement

Given

A generating circle of diameter 50 mm rolls internally without slipping inside a directing circular arc of radius 85 mm for one revolution.

- Construct the hypocycloid.
- Draw a tangent and normal at a point 50 mm from the centre of the directing circle.
- Retain the construction geometry in the final drawing.

Calculated Data

$$R = 85 \text{ mm},$$

$$OC_0 = R - r = 85 - 25 = 60 \text{ mm},$$

$$\angle AOB = 360^\circ \frac{r}{R} = 360^\circ \frac{25}{85} = 105.882353^\circ,$$

$$\Delta\theta = \frac{105.882353^\circ}{12} = 8.823529^\circ,$$

$$r = \frac{50}{2} = 25 \text{ mm},$$

$$AC_0 = 25 \text{ mm},$$

Generating-circle division = 30° .

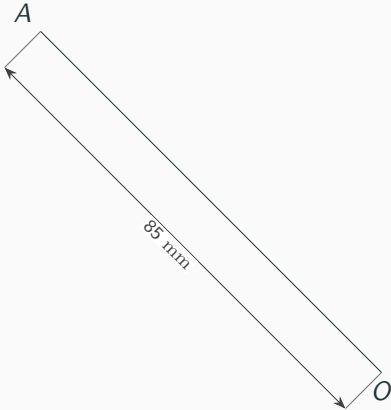
$$C(\theta) = 60(\cos(135^\circ - \theta), \sin(135^\circ - \theta)),$$

$$P(\theta) = C(\theta) + 25(\cos(135^\circ + 2.4\theta), \sin(135^\circ + 2.4\theta)).$$

- $2.4 = (R - r)/r = 60/25$ is the rolling-circle rotation factor.
- The initial generating-circle divisions are numbered clockwise.
- Concentric arcs are drawn with O as centre through those divisions.
- Each C_i , with radius 25 mm, cuts the corresponding concentric arc at P_i .

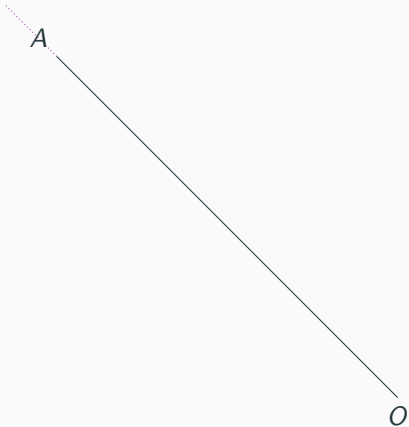
Step 1: Draw OA

$OA = 85 \text{ mm.}$



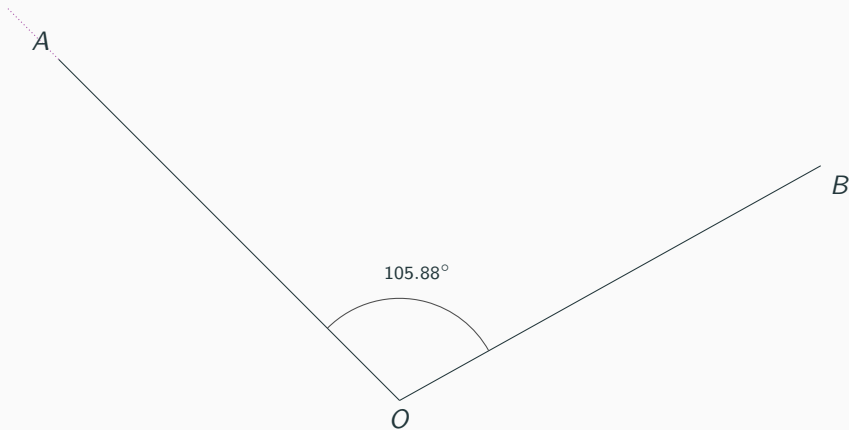
Step 2: Extend OA

Extend OA .

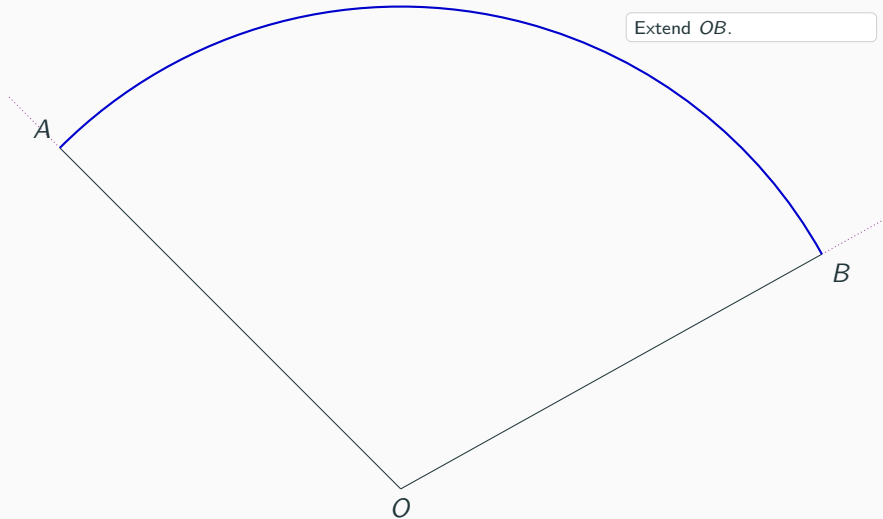


Step 3: Draw OB at 105.88°

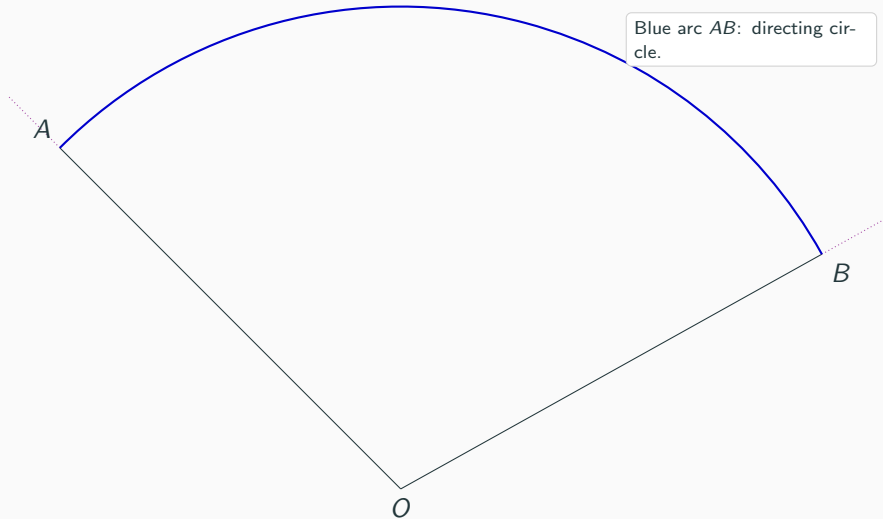
$$\angle AOB = 105.88^\circ.$$



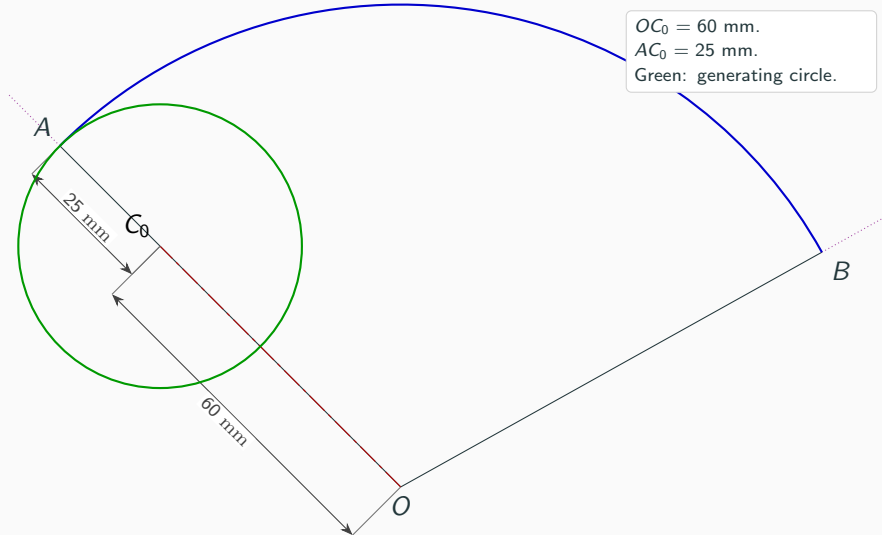
Step 4: Extend OB



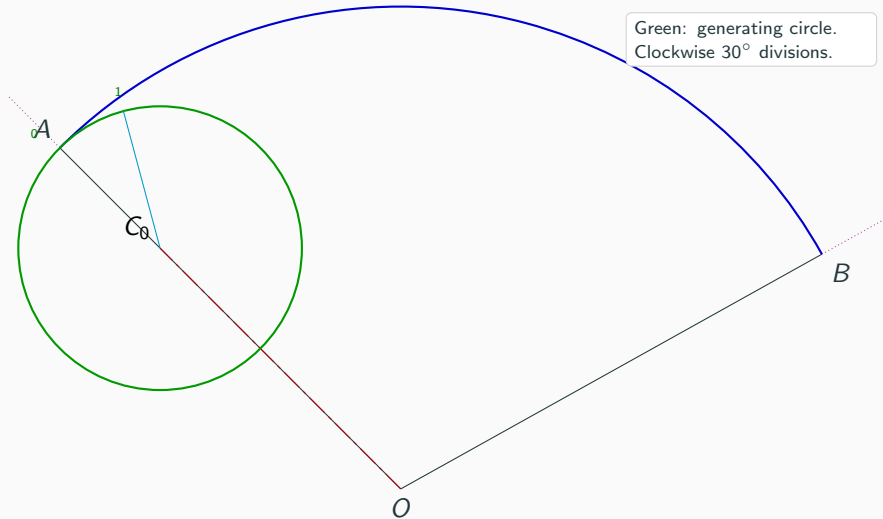
Step 5: Draw the directing arc AB



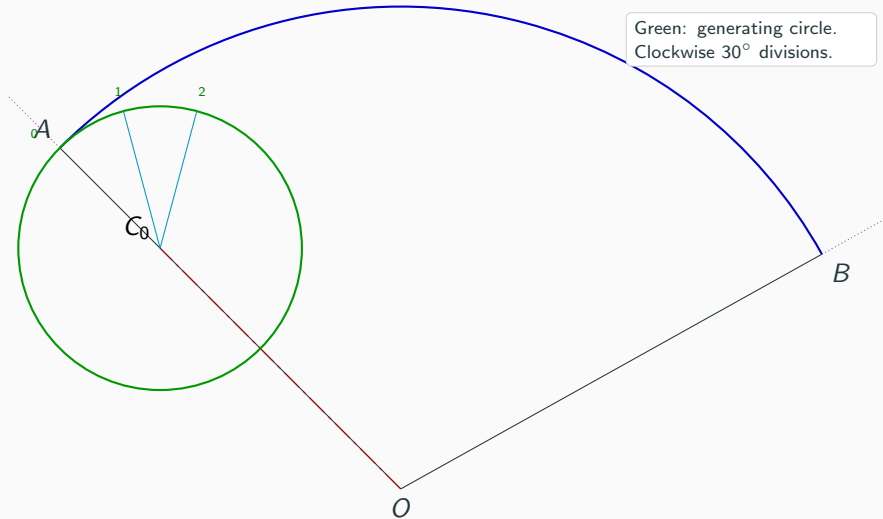
Step 6: Locate C_0 and the generating circle



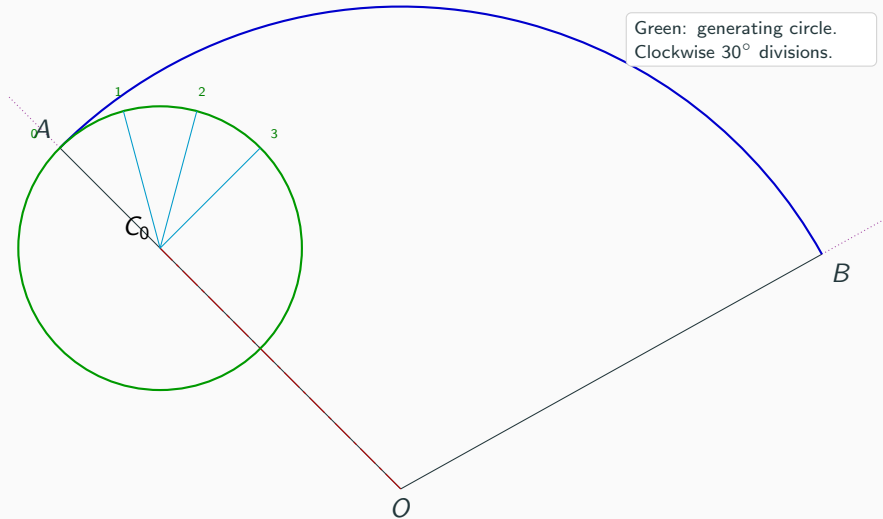
Step 7: Divide the generating circle into 12 equal parts



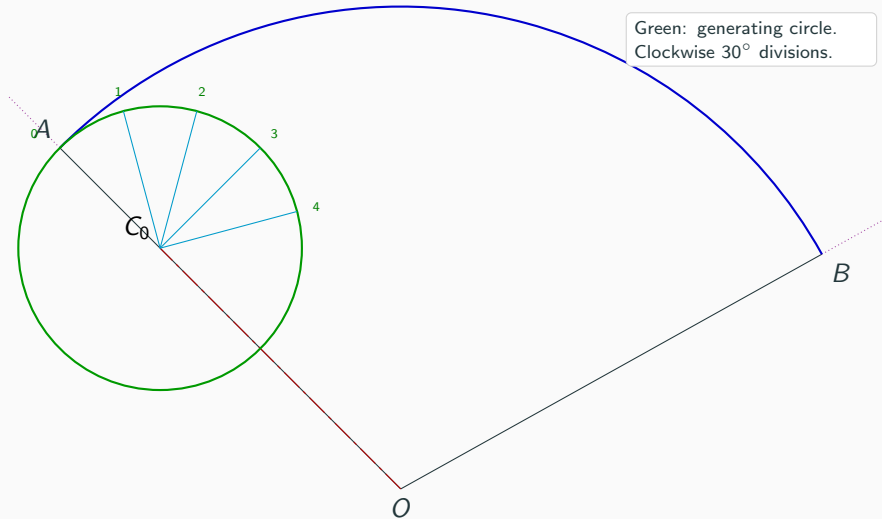
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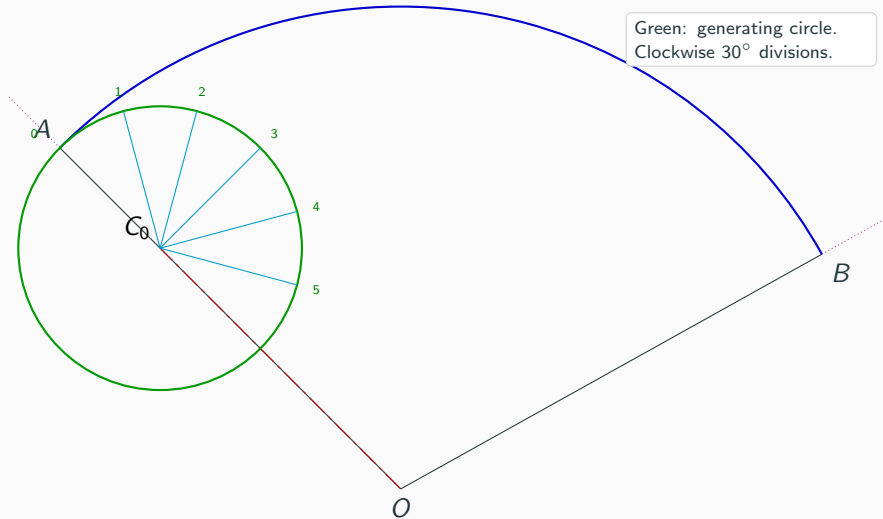
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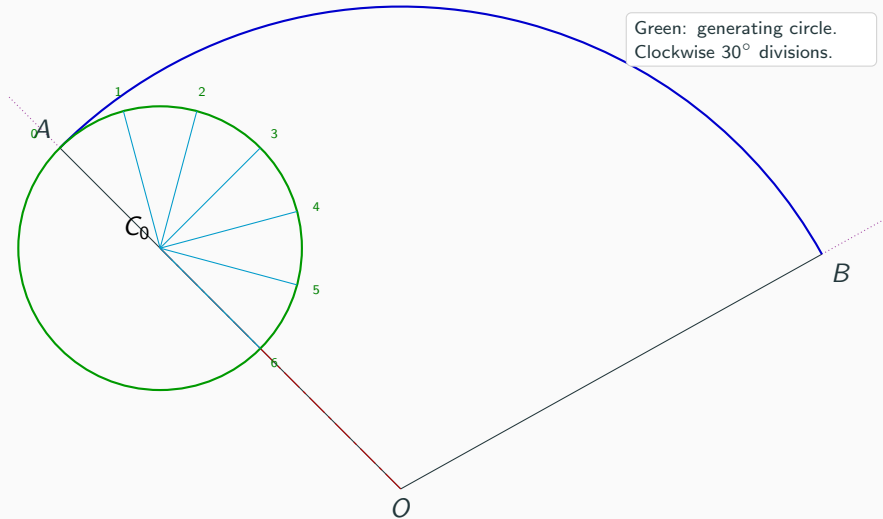
Step 7: Divide the generating circle into 12 equal parts



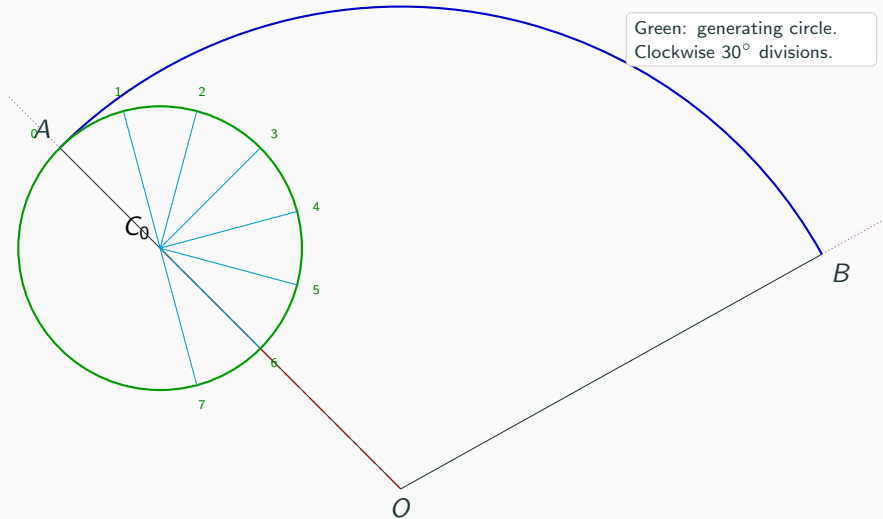
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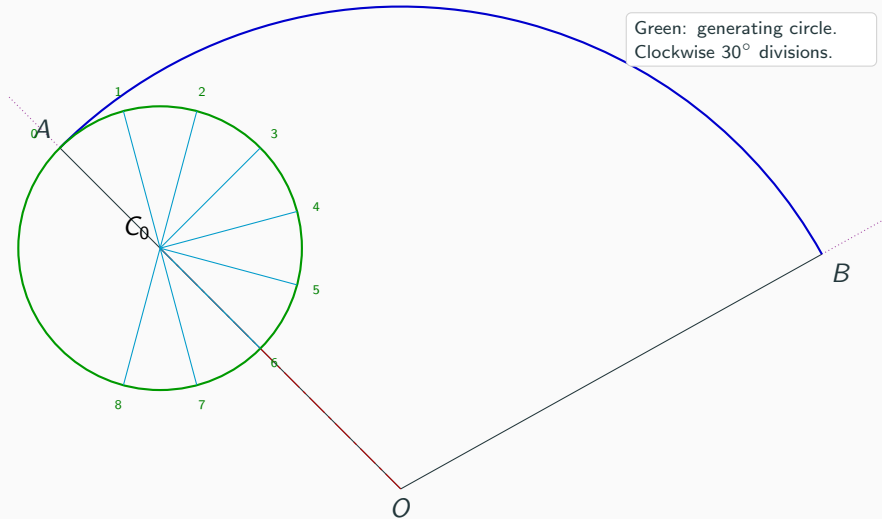
Step 7: Divide the generating circle into 12 equal parts



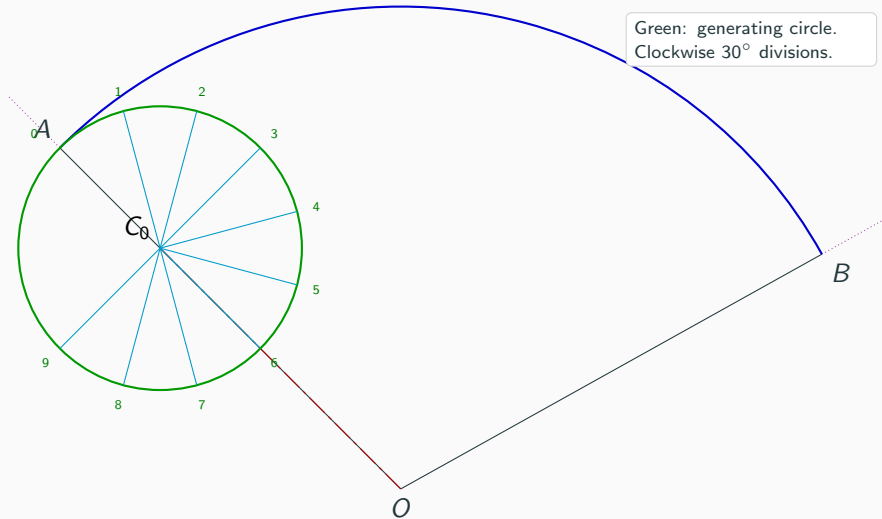
Step 7: Divide the generating circle into 12 equal parts



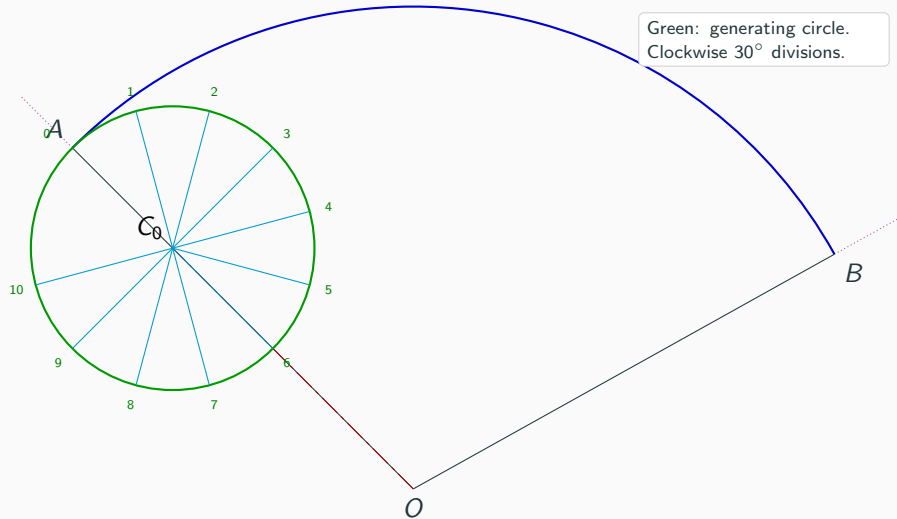
Step 7: Divide the generating circle into 12 equal parts



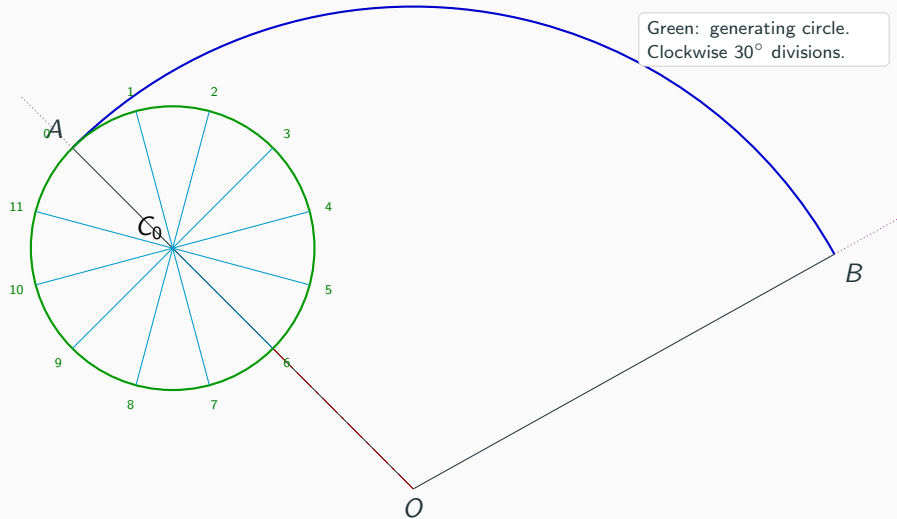
Step 7: Divide the generating circle into 12 equal parts



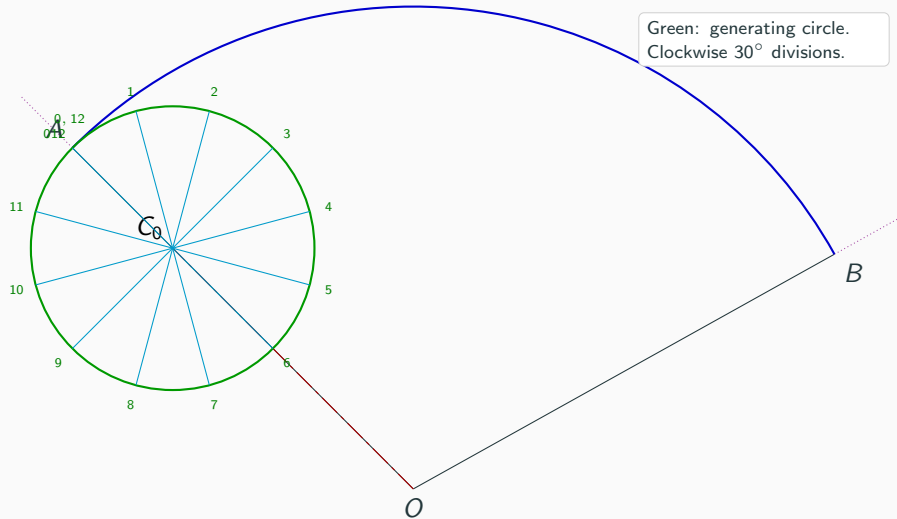
Step 7: Divide the generating circle into 12 equal parts



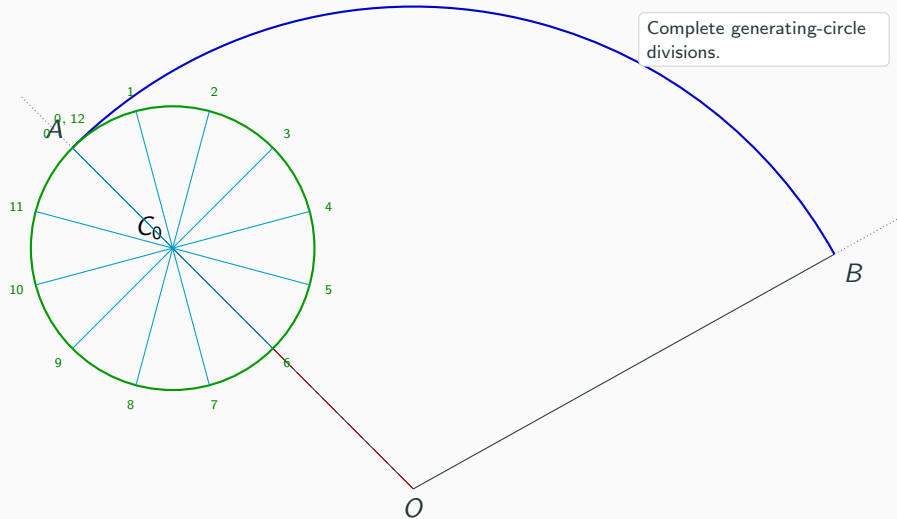
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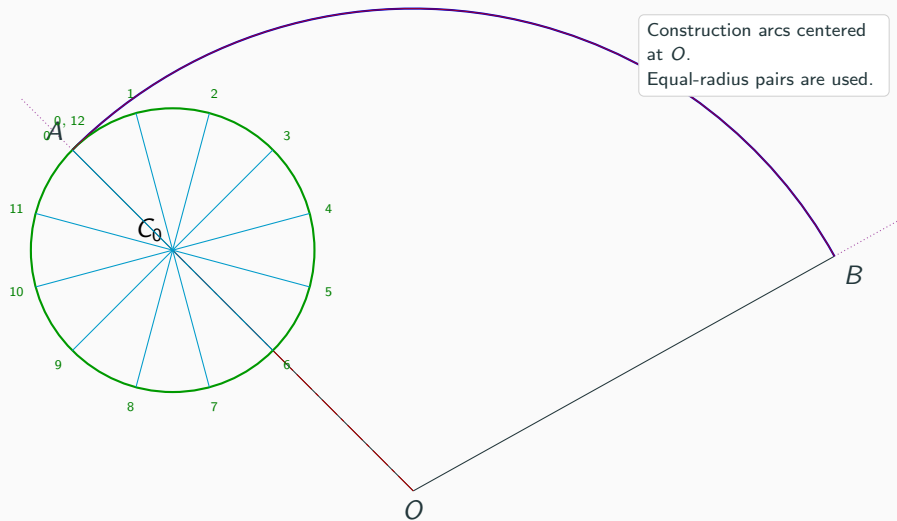
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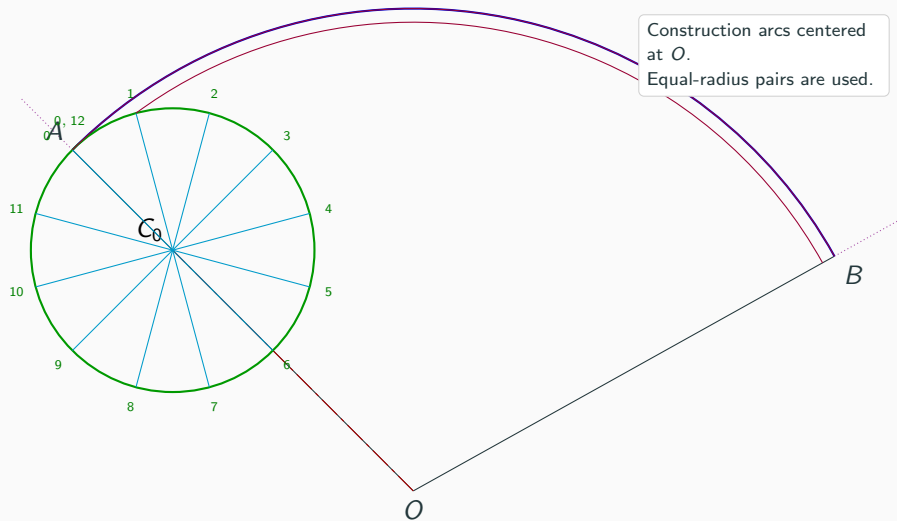
Step 8: Complete the generating-circle divisions



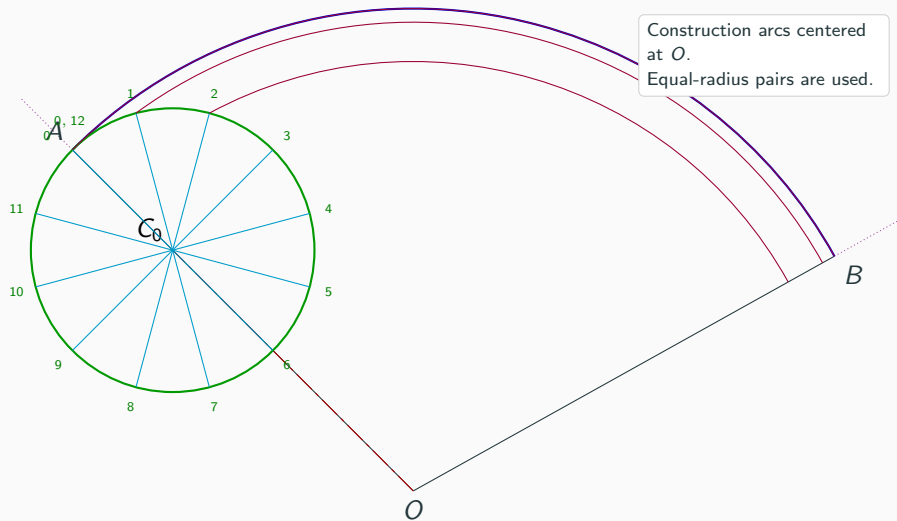
Step 9: Draw concentric construction arcs



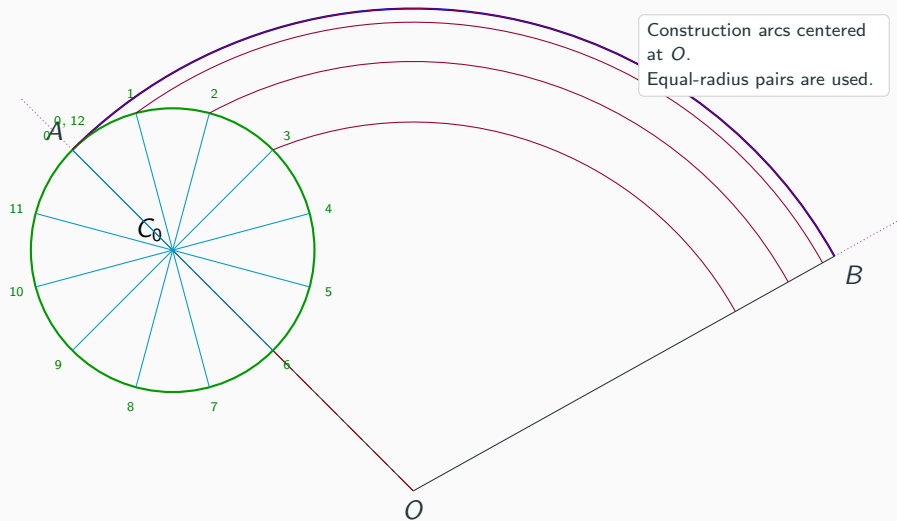
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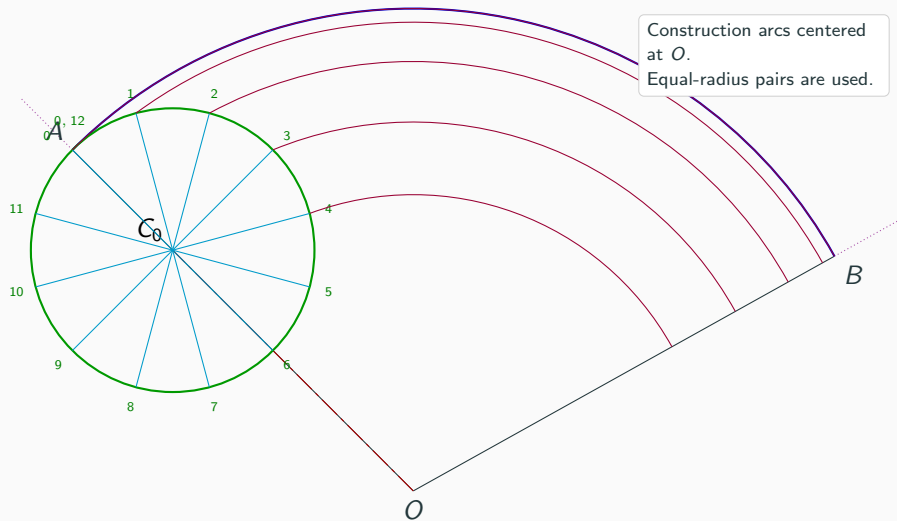
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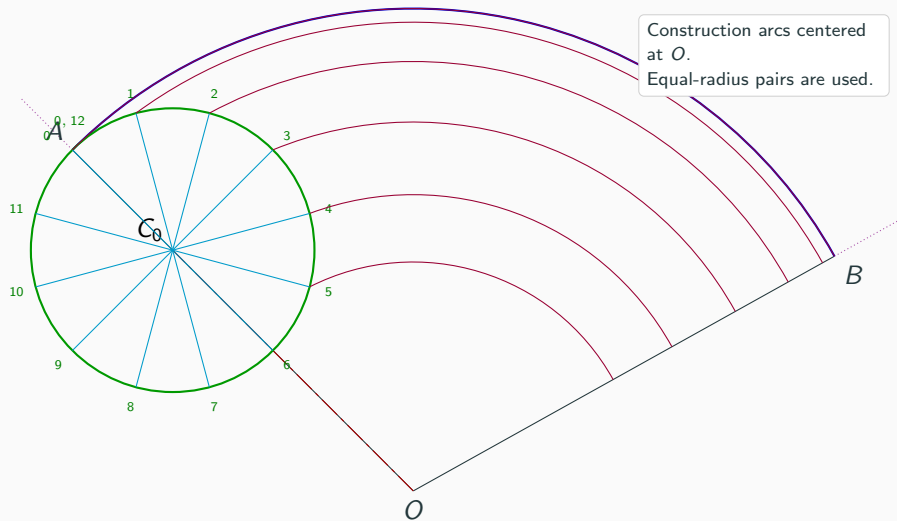
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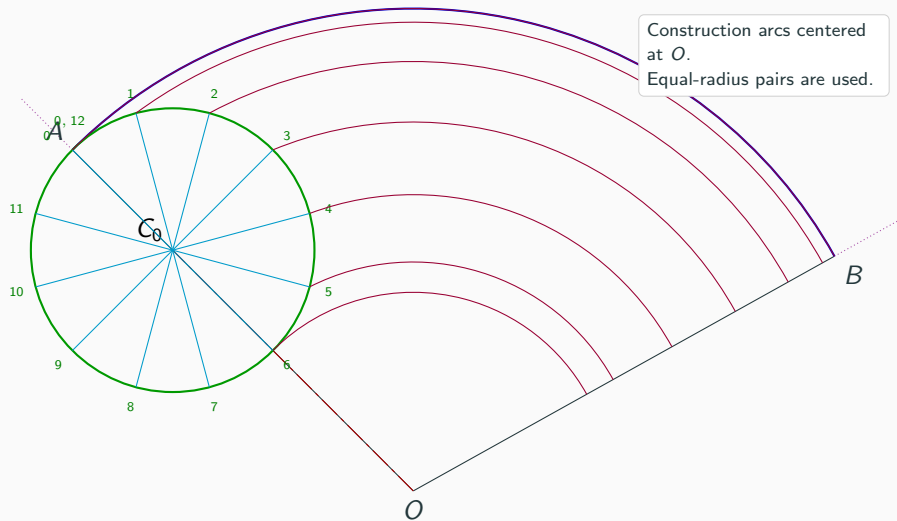
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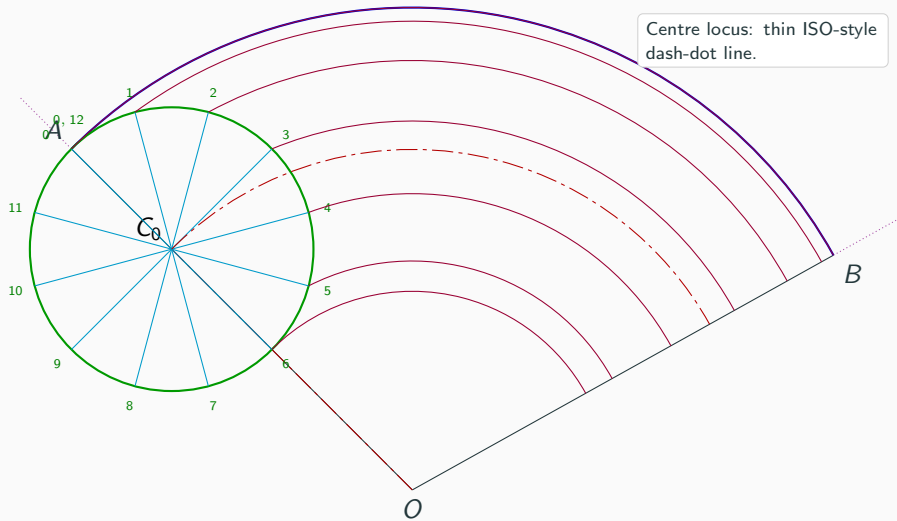
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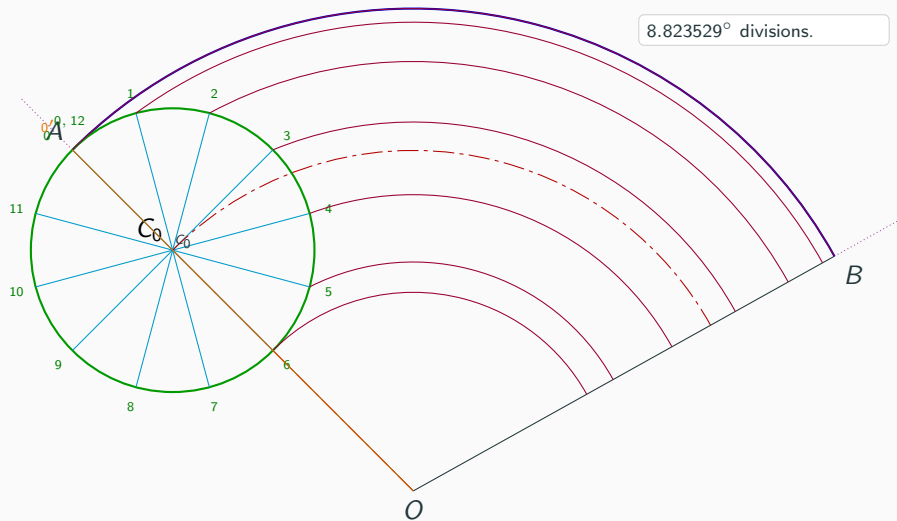
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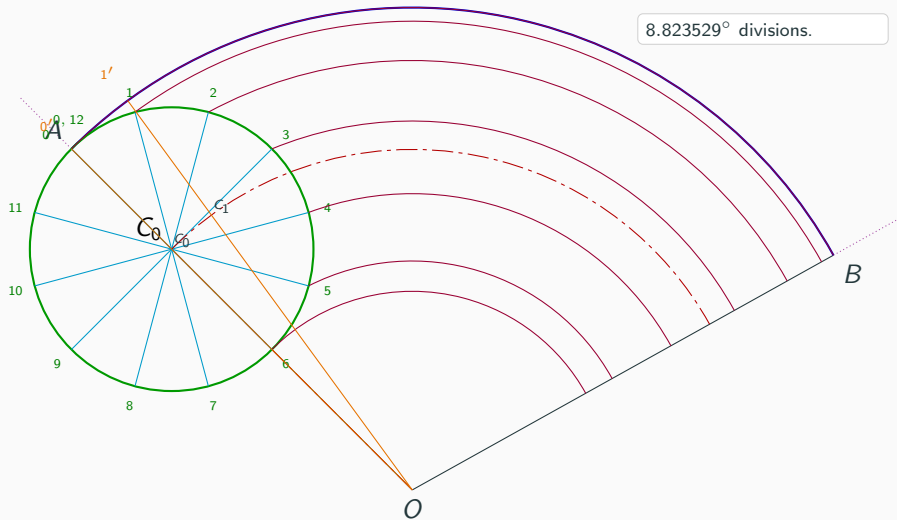
Step 10: Draw the centre arc through C_0



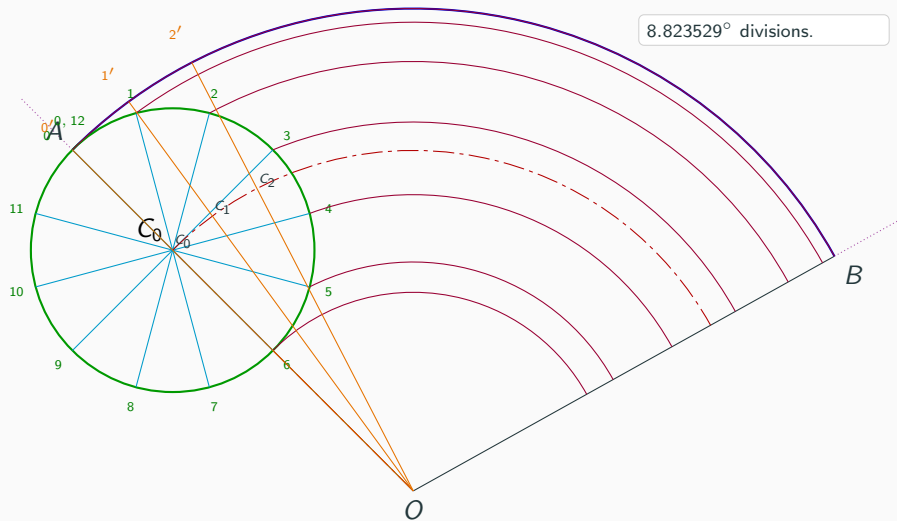
Step 11: Divide $\angle AOB$ into 12 equal parts



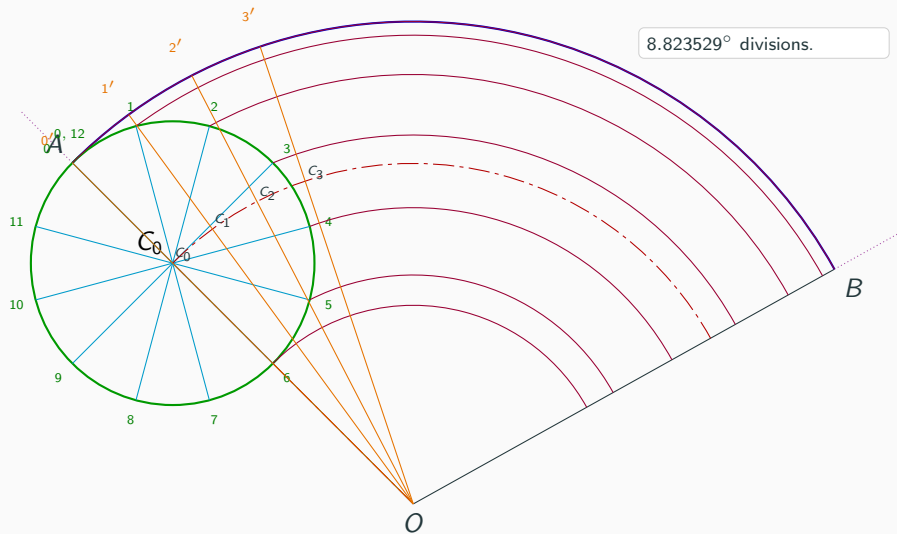
Step 11: Divide $\angle AOB$ into 12 equal parts



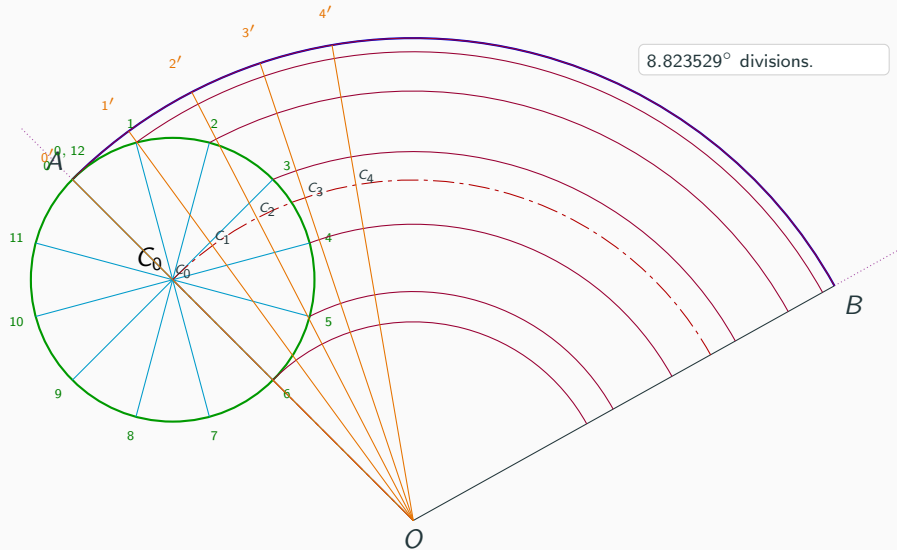
Step 11: Divide $\angle AOB$ into 12 equal parts



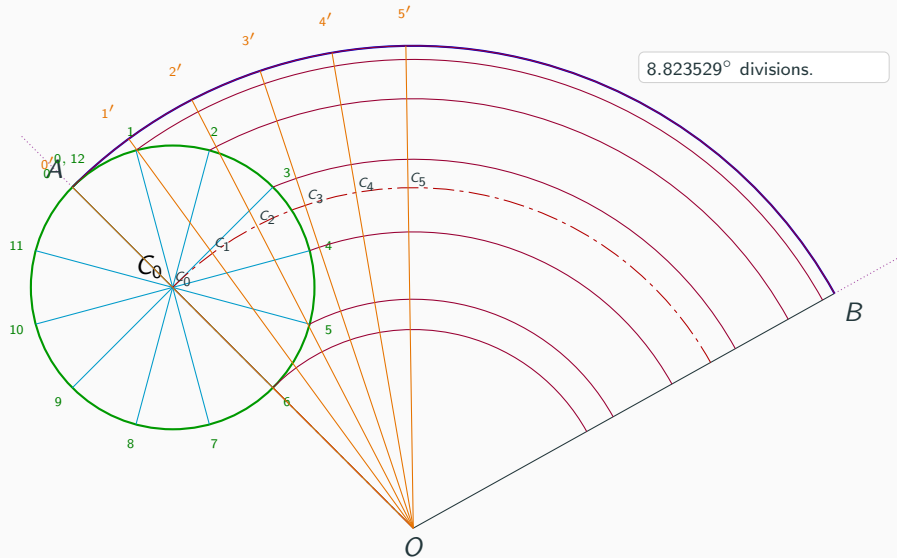
Step 11: Divide $\angle AOB$ into 12 equal parts



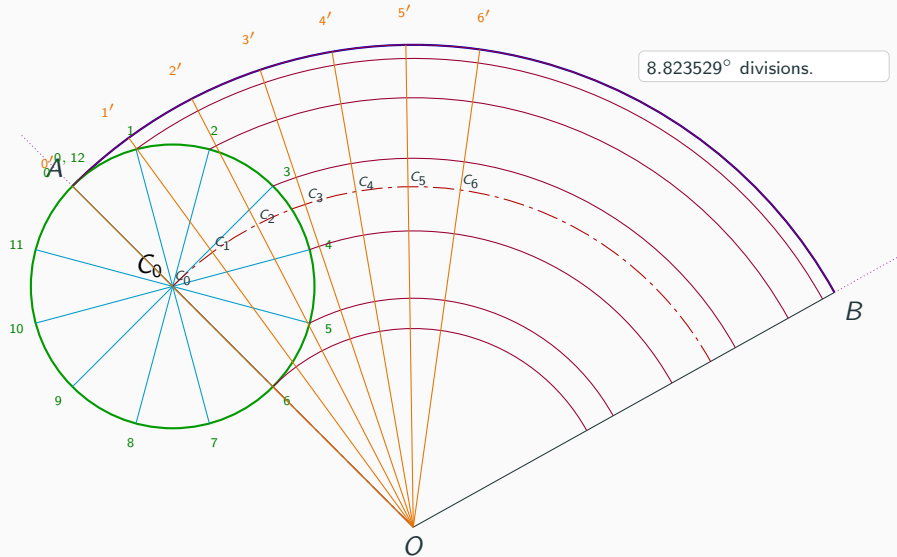
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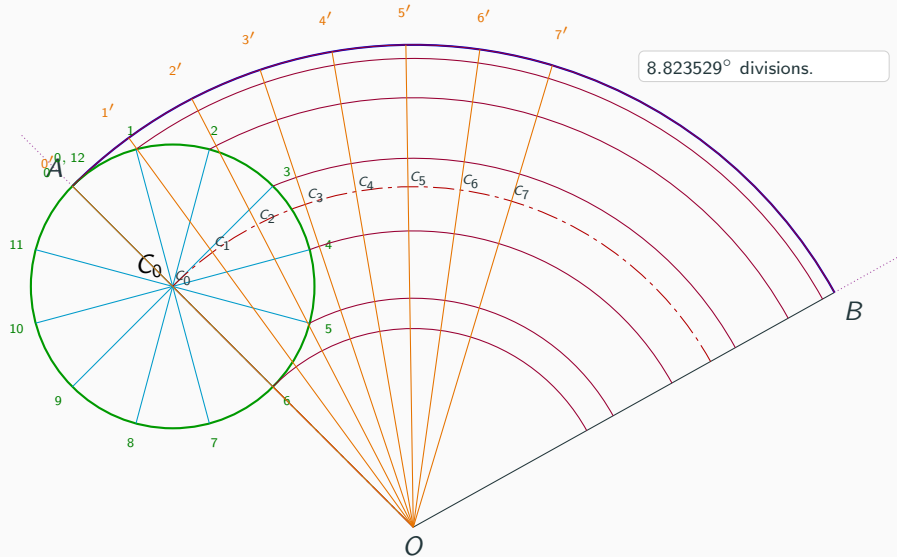
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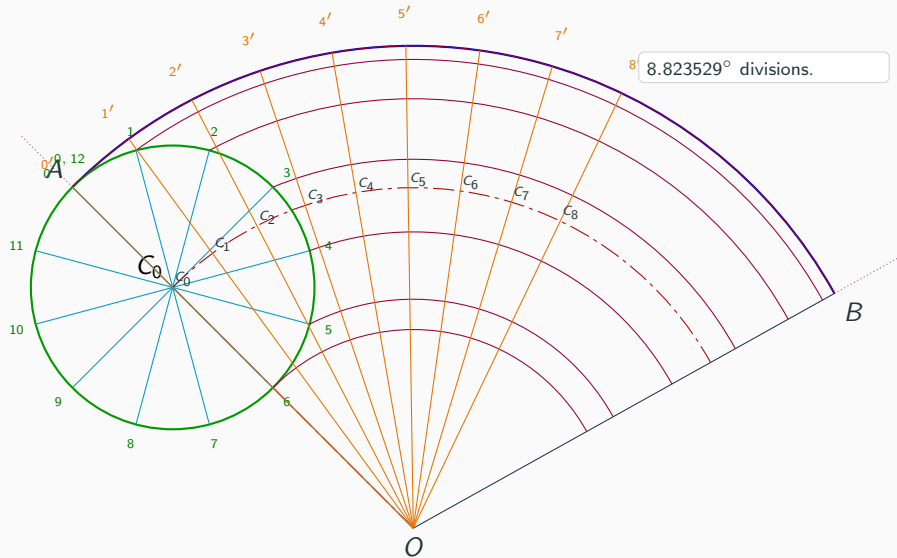
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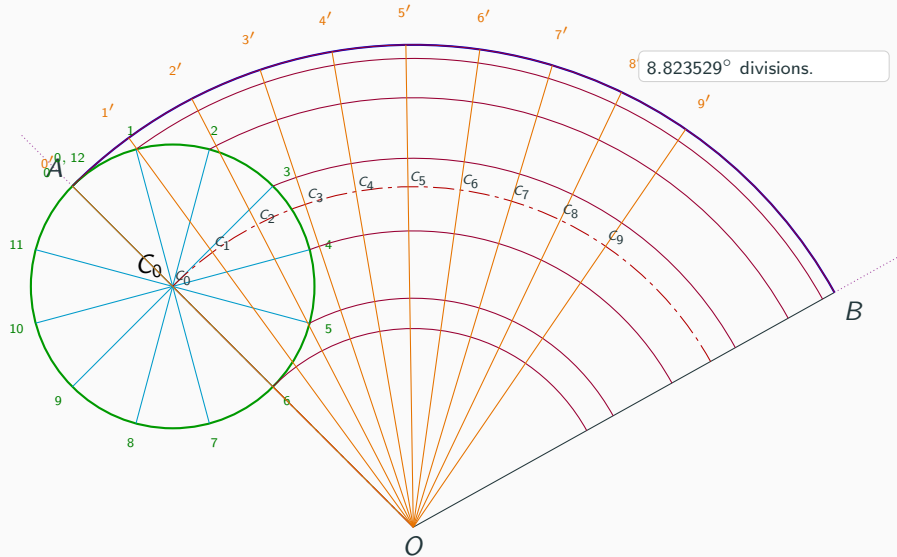
Step 11: Divide $\angle AOB$ into 12 equal parts



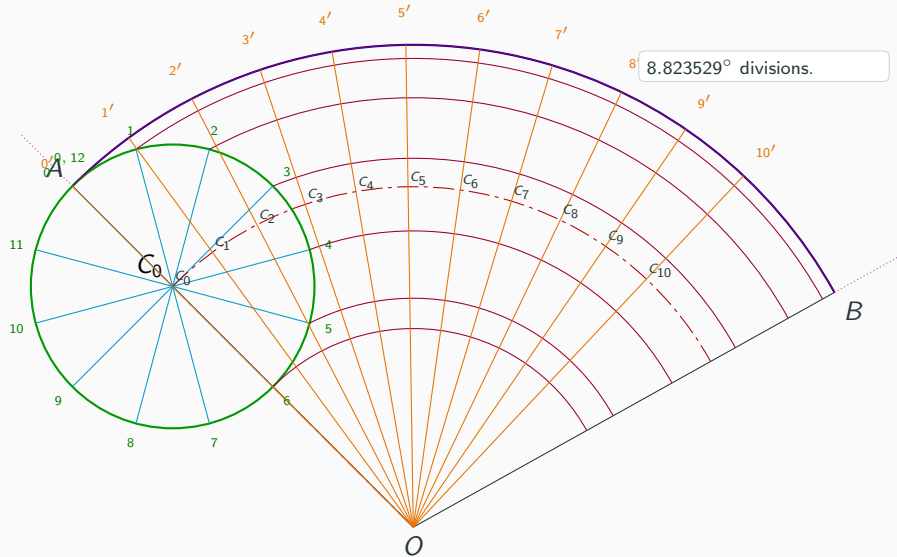
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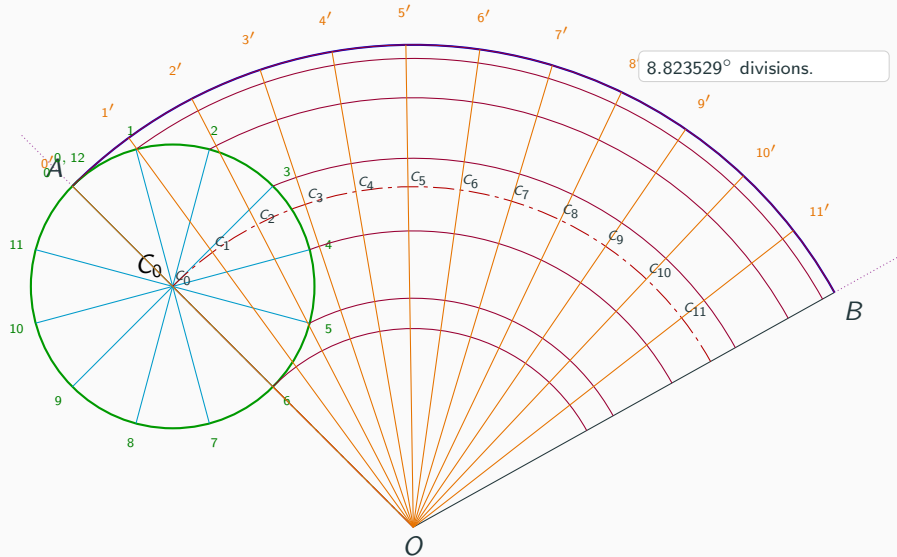
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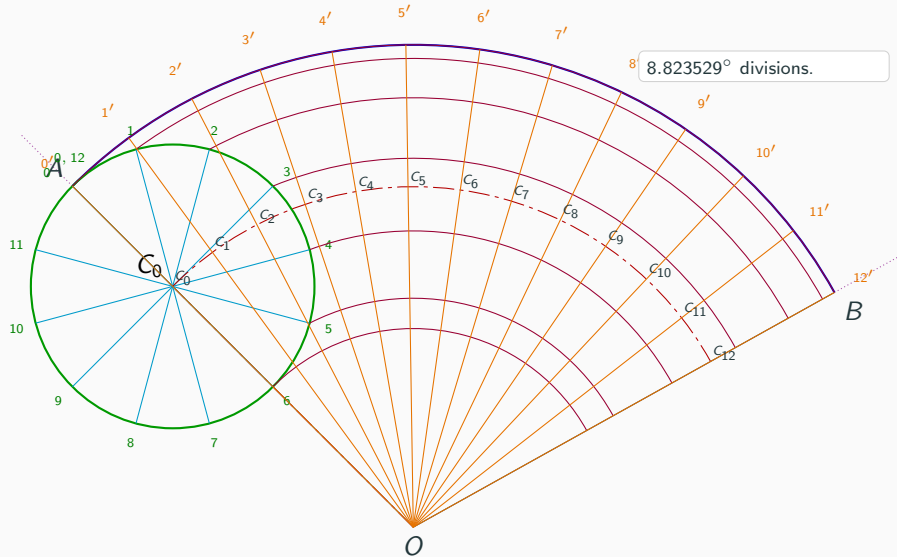
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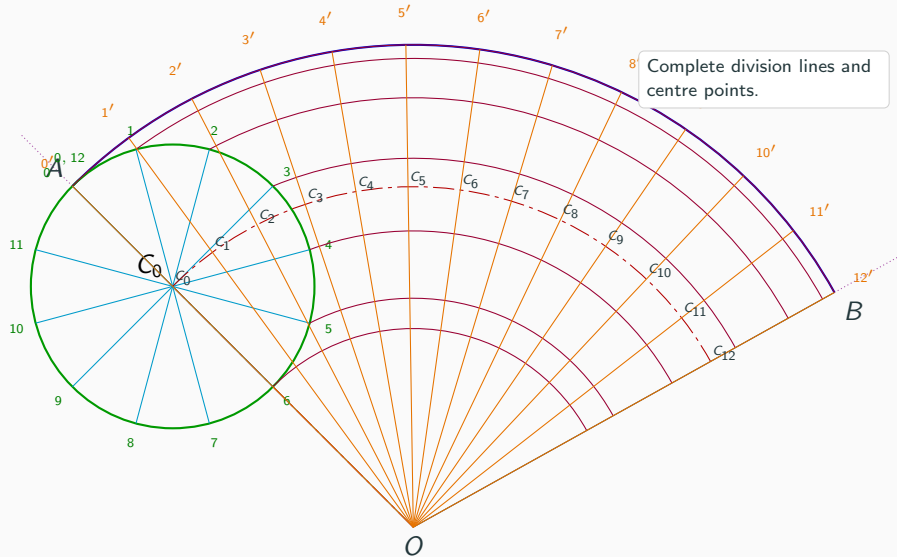
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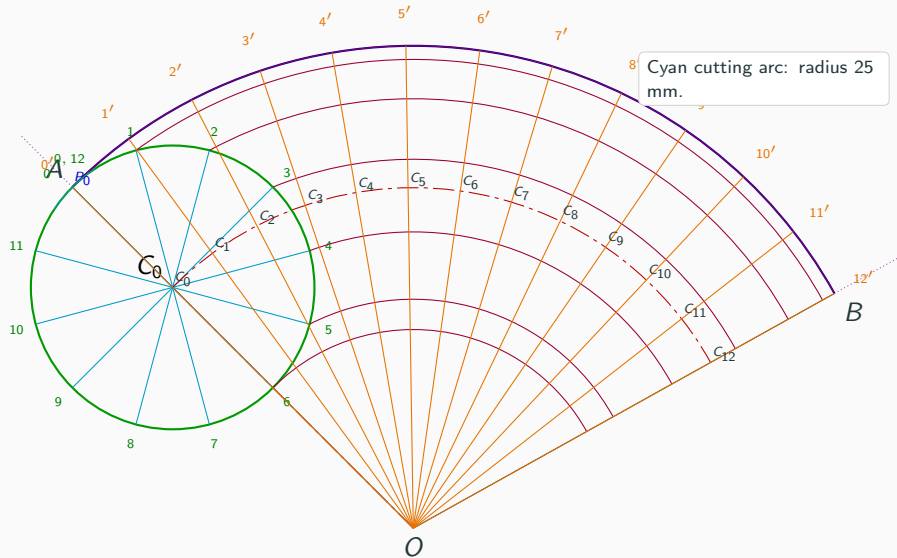
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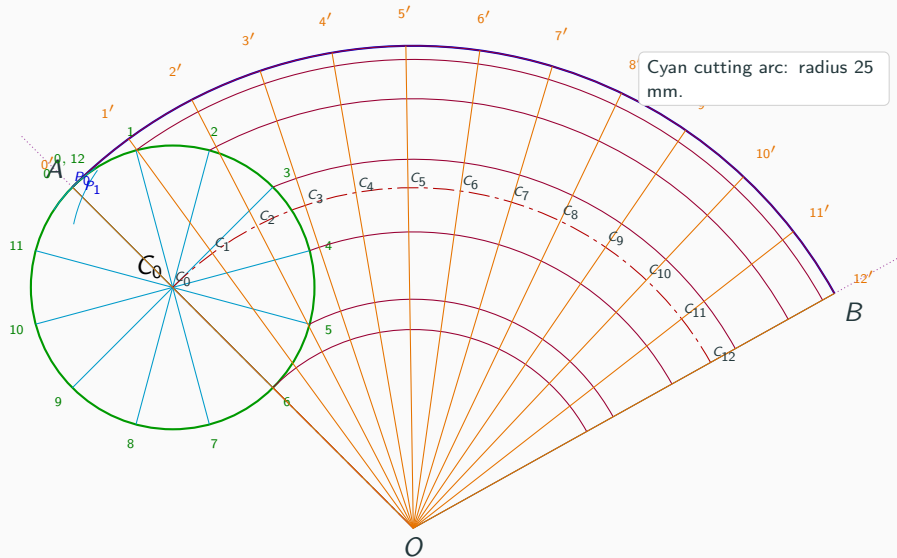
Step 12: Complete the angular divisions and centre points



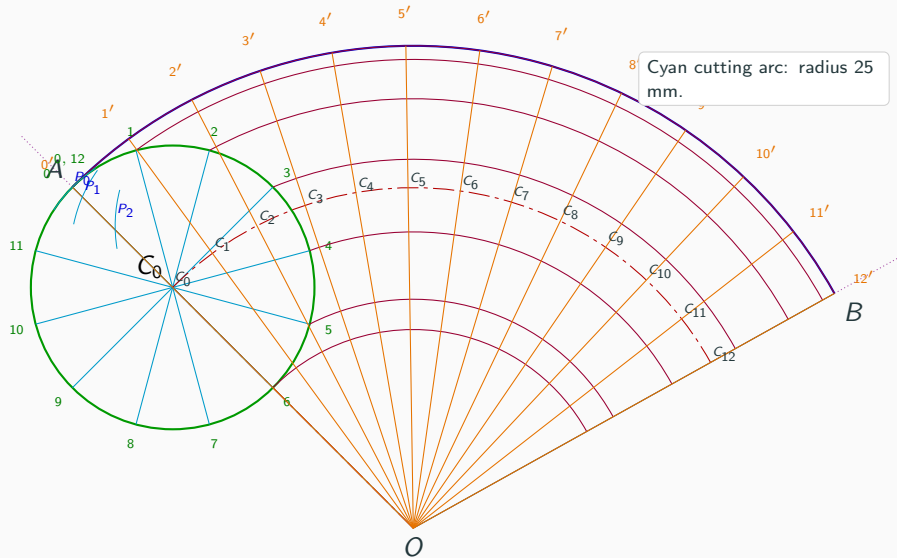
Step 13: Draw the cutting arc and locate P_i



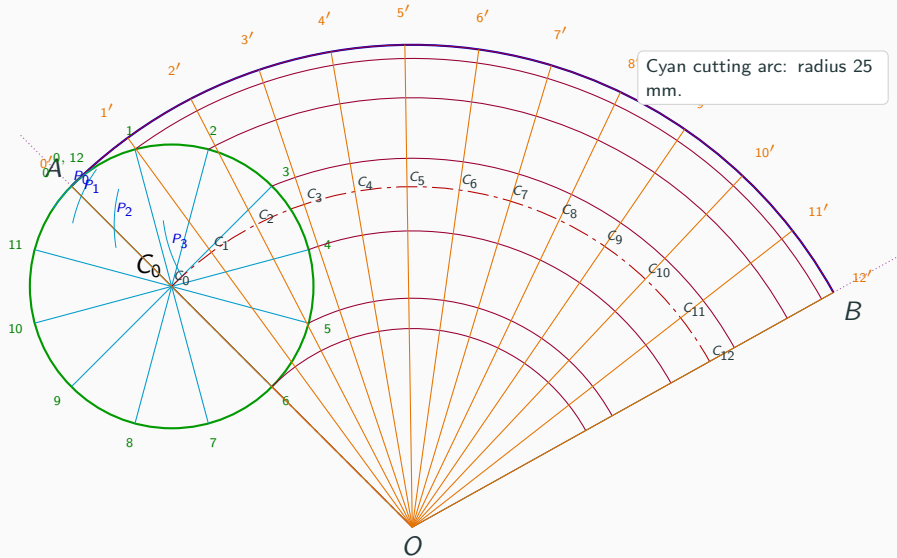
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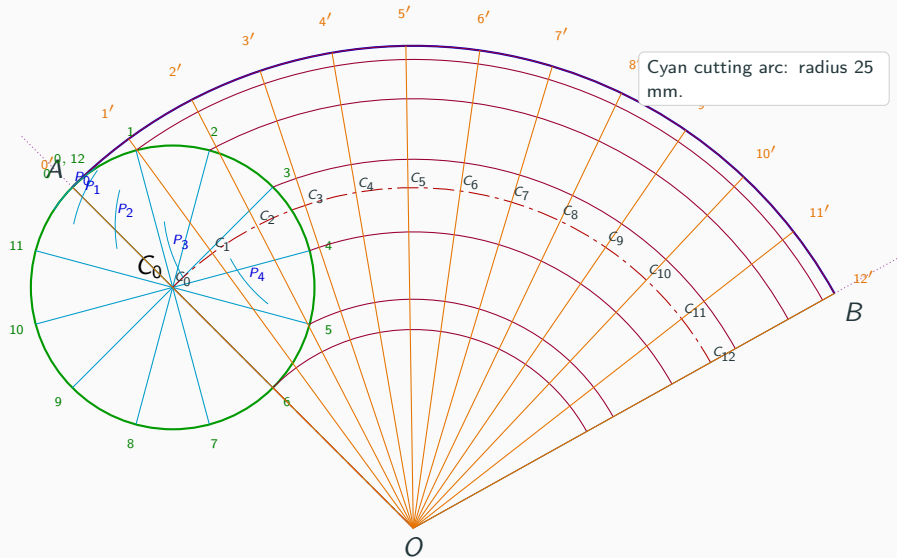
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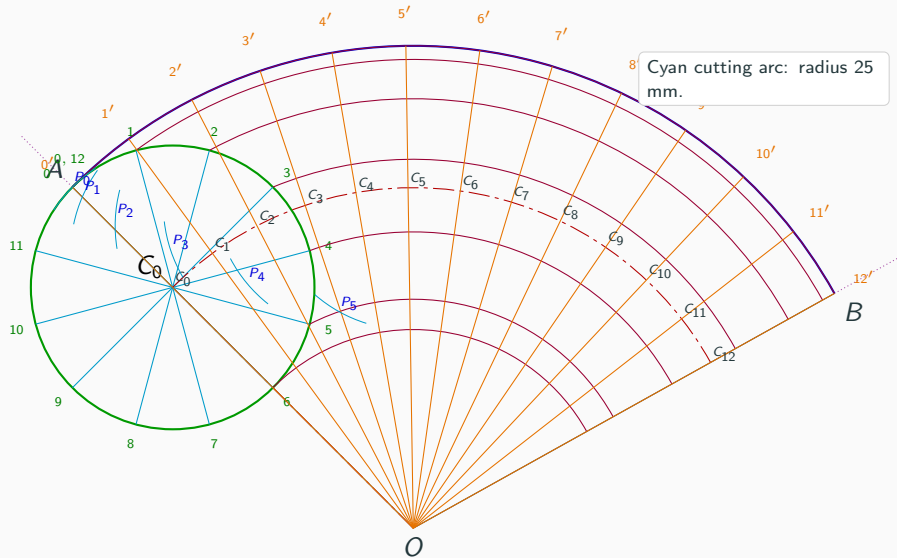
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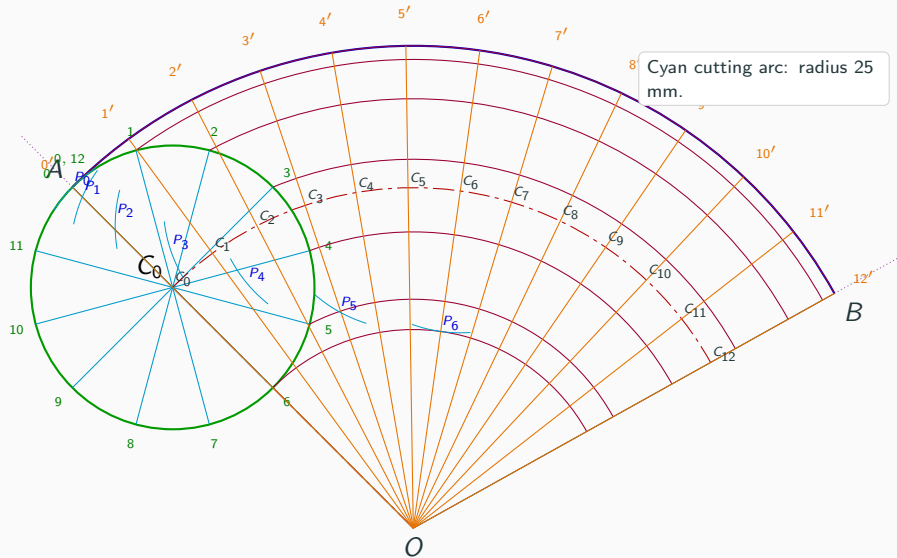
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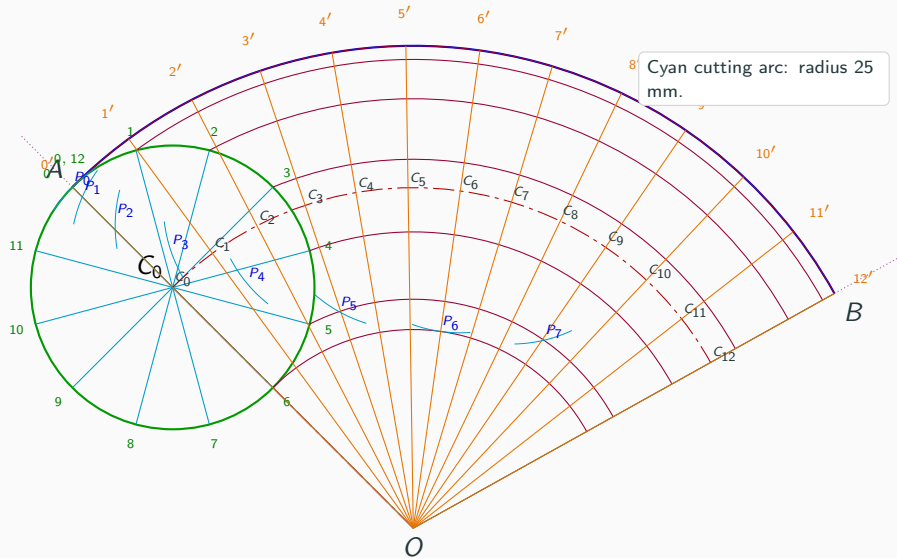
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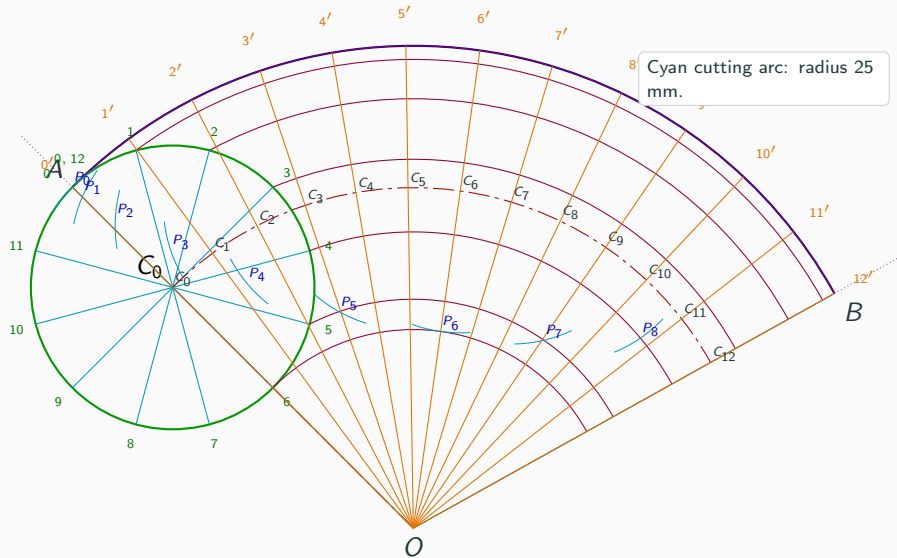
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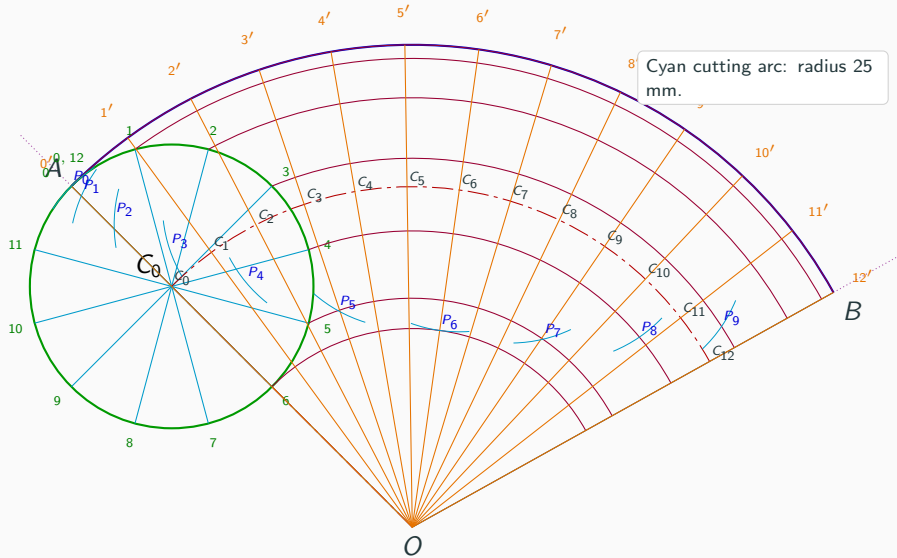
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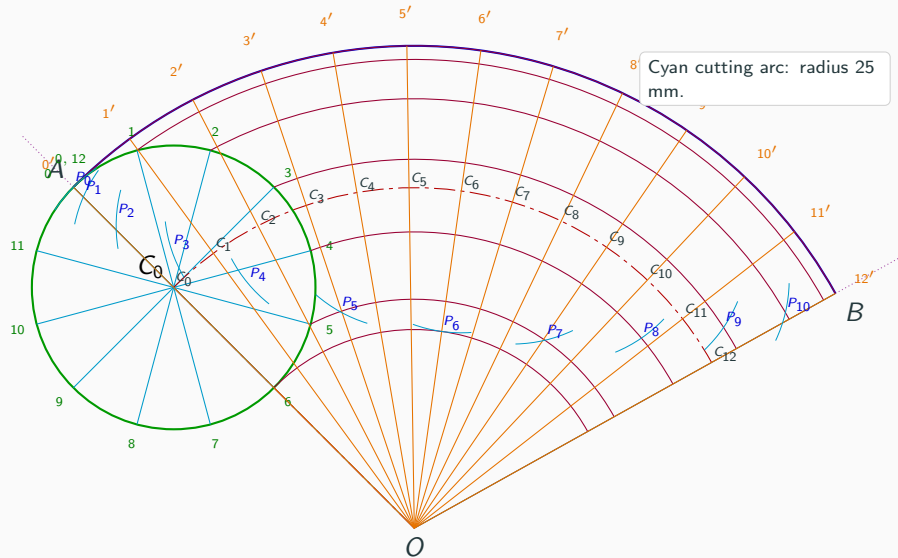
Step 13: Draw the cutting arc and locate P_i



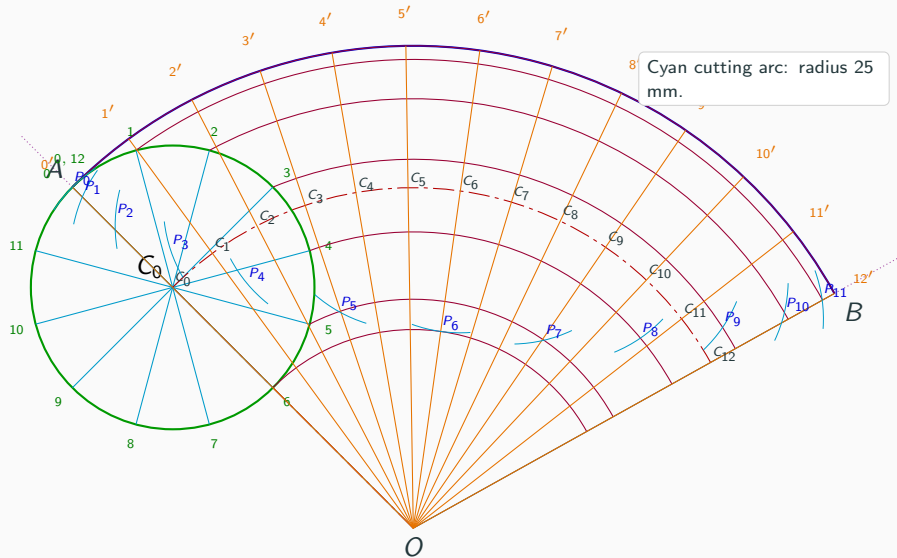
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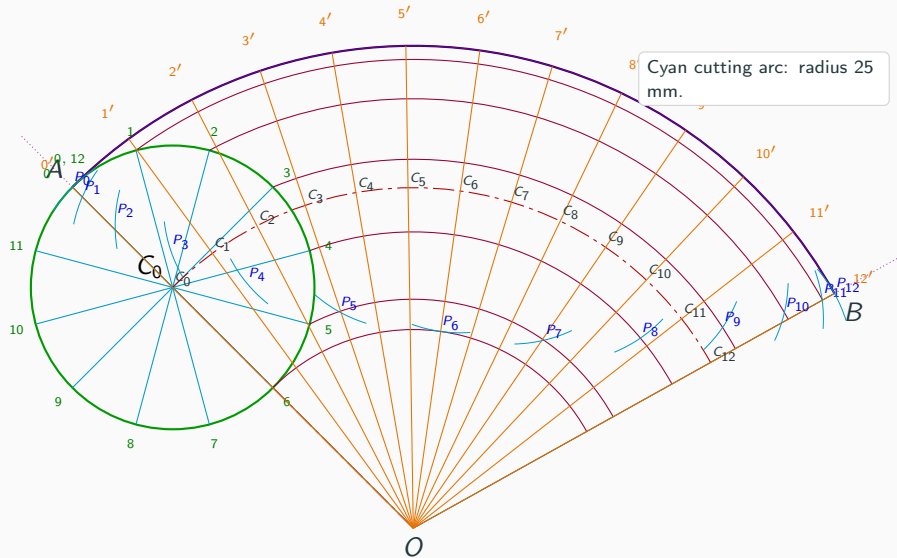
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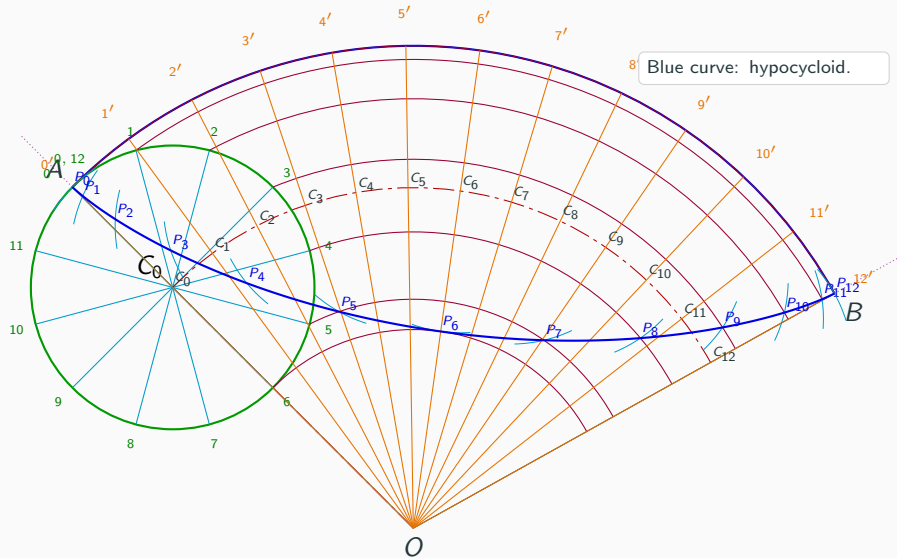
Step 13: Draw the cutting arc and locate P_i



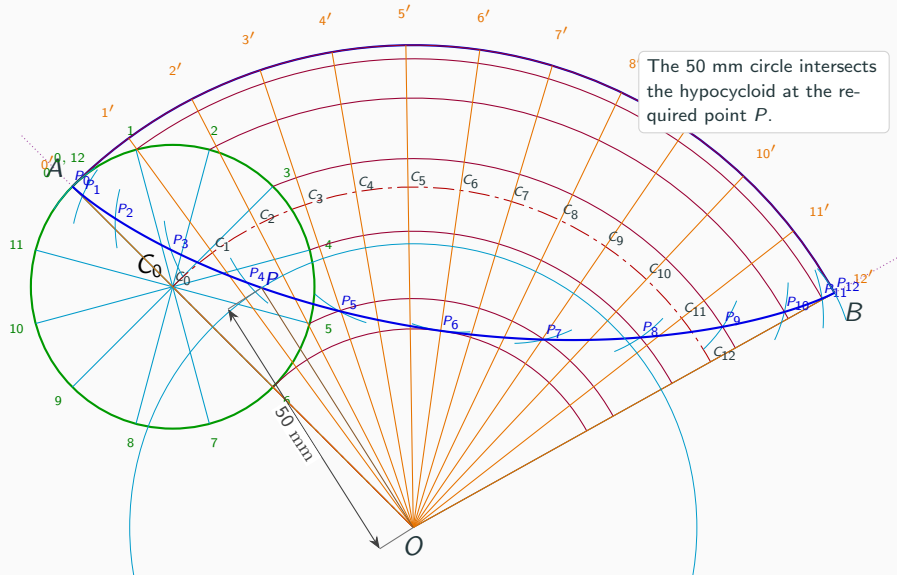
Step 13: Draw the cutting arc and locate P_i



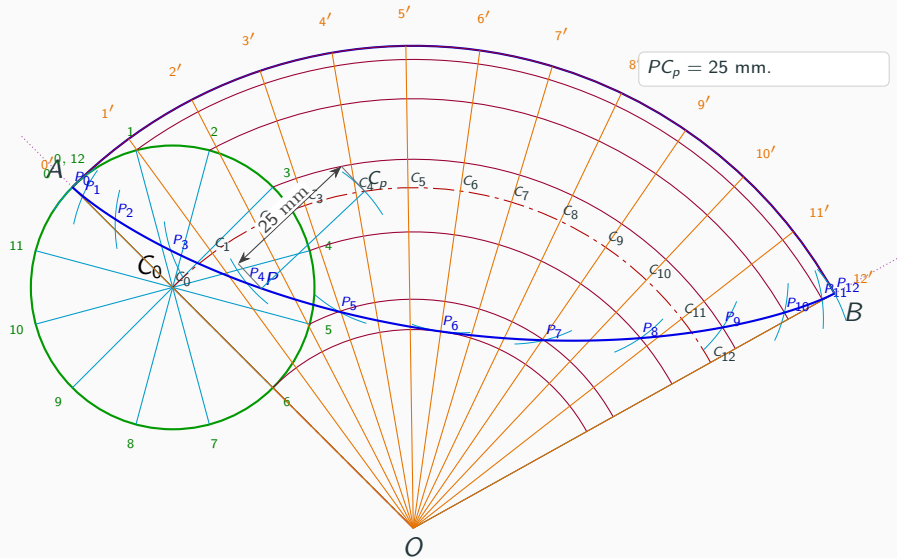
Step 14: Draw the hypocycloid



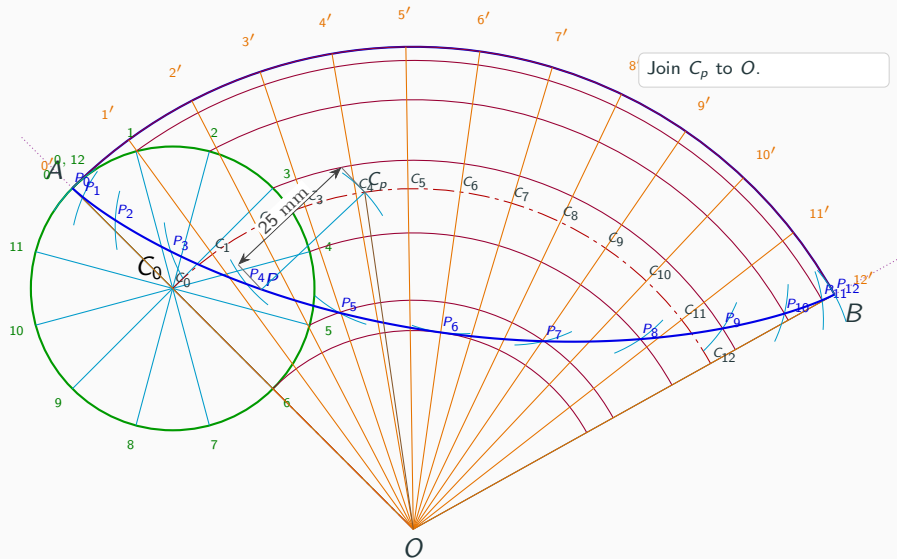
Step 15: Locate P such that $OP = 50$ mm



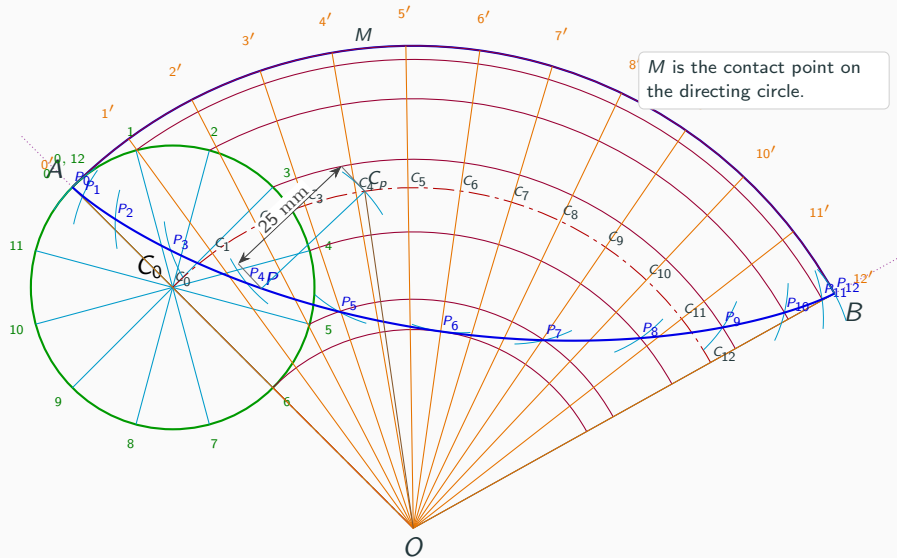
Step 16: Draw the 25 mm radius to C_p



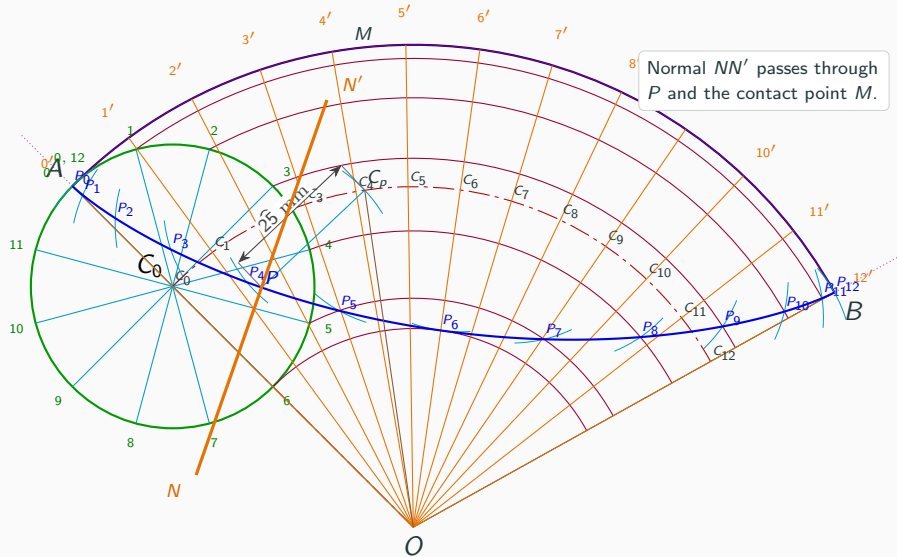
Step 17: Join C_p to O



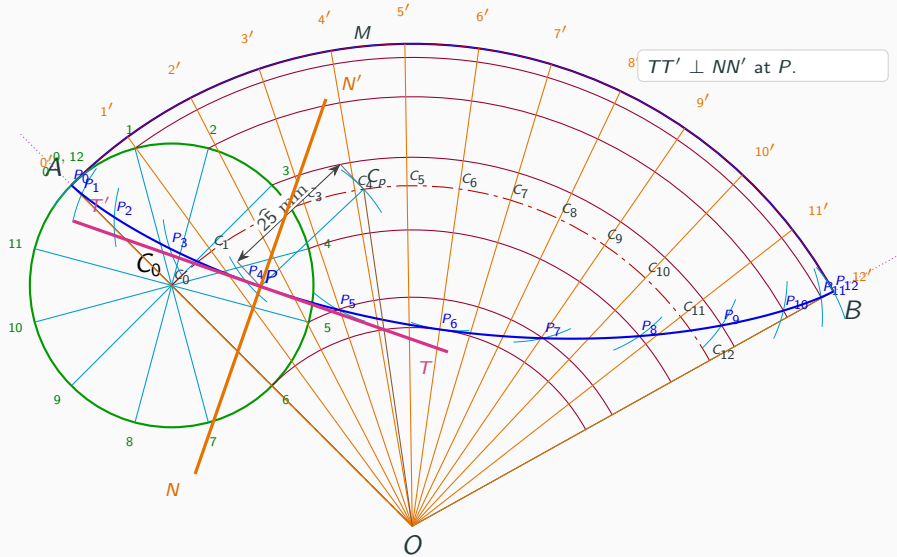
Step 18: Mark M on the directing circle



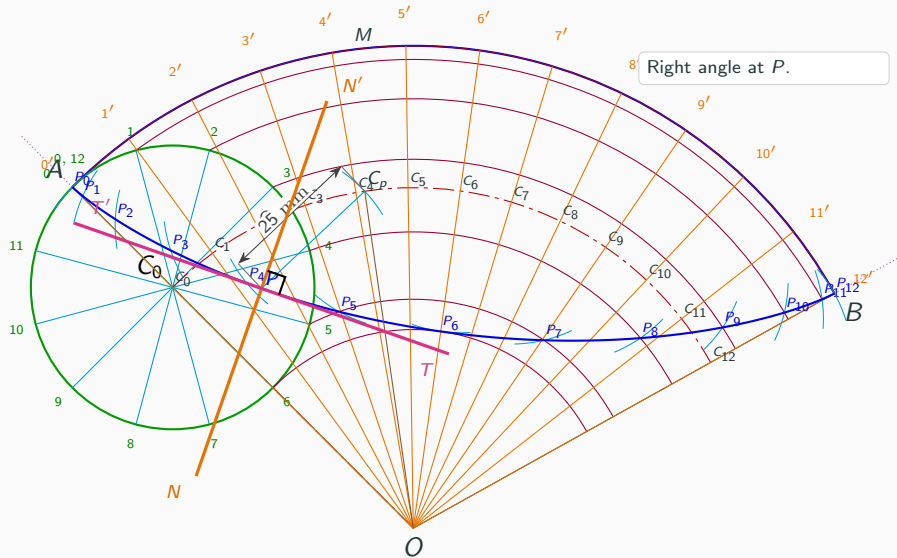
Step 19: Draw the normal NN' through P



Step 20: Draw the tangent TT' perpendicular to NN'



Step 21: Show the right angle at P



Step 22: Final completed hypocycloid construction

